

Musaad Hydary

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Computer Engineer specializing in product and UI/UX engineering, with hands-on full-stack experience across embedded systems, mobile applications, and consumer hardware-software integration.

EDUCATION

Queen's University

Kingston, ON

Bachelor of Applied Science in Computer Engineering

September 2021 – April 2026

- Coursework: Operating Systems, Algorithms, Database Management, Distributed Systems, Performance Analysis

SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, Java, HTML5, CSS, SQL, Verilog, RISC-V Assembly

Technologies: React, React Native, Node.js, GraphQL, PostgreSQL, Prisma, Redux, Supabase, TailwindCSS, Tauri

Tools: Git, Docker, Kubernetes, Vercel, XCode, Expo, Datadog, Figma, Blender, Arduino, Roboflow, Microsoft Office

ENGINEERING EXPERIENCE

Software Engineer

May 2024 – August 2025

ARENA Exchange

Toronto, ON

- Architected scalable backend APIs with **Node.js**, **GraphQL**, and **Prisma**, integrating **Mixpanel**-powered real-time market alerts that increased user engagement by **24%**
- Built custom profiling tools and implemented **Datadog APM** monitoring to diagnose critical bottlenecks, improving iOS app performance from **49 FPS to 58 FPS (20% increase)** and doubling response times
- Redesigned UI components to enhance mobile app responsiveness and accessibility across the platform

Systems Engineer

September 2023 – April 2025

Queen's AutoDrive Design Team

Kingston, ON

- Engineered lighting components in **C++** with **CAN bus** protocols and optimized data structures, reducing message latency for real-time **ROS** communications to enable safer pedestrian interaction
- Labeled datasets using **Roboflow** to improve object detection accuracy for autonomous vehicle perception
- Led a team of 4 and presented at SAE World Congress 2024 and 2025 on autonomous vehicle safety, securing **2nd place** among 10+ international teams and contributing to industry accessibility standards

PROJECTS

Glucowatch: Hybrid Smartwatch for Diabetes Management

- Built a **React Native** app with 8+ custom components implementing **Bluetooth Low Energy** protocols, **HealthKit** integration, and a three-tier data fallback system to enable continuous, reliable glucose tracking

UVCAT: Image Processing & UV Mapping Platform

- Built a full image-processing pipeline with custom dithering algorithms and **3D UV mapping** visualization, backed by a **GraphQL/Prisma** API and community texture-pack marketplace with authentication

PIXEL-V: Browser-Based RISC-V CPU Emulator

- Designed and deployed a complete **RISC-V** CPU emulator with a custom assembler, instruction decoder, and memory-mapped graphics runtime, running entirely client-side in the browser using **TypeScript**

Macropad Configurator

- Developed a native macOS desktop app with **Tauri** and **Rust** implementing low-latency **HID**-level event tapping via a multi-process architecture, enabling real-time macro key detection, remapping, and configuration

EXTRACURRICULAR LEADERSHIP

Co-President

July 2025 – June 2026

Queen's Music Production Club (QWave)

Kingston, ON

- Led 12+ studio jam sessions and production workshops, grew membership from 150 to 250+ students, and managed DAW software distribution for the club community

Public Speaker

September 2024 – August 2025

Toastmasters International at High Park Speakers

Toronto, ON

- Won **1st place** in a speech competition among 8+ Toronto clubs and earned **Triple Crown award** for completing communication and leadership tracks while mentoring club members